

Cartesian tensorization of the graph entropy constant and the exact constant of weighted Boolean cubes

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Abstract

For every pair of finite weighted graphs, equipped with uniform vertex measures and standard Cartesian-product conductances, we prove

$$\kappa(G \square H) = \min\{\kappa(G), \kappa(H)\}.$$

The spectral gap has the same product rule. Consequently the equality $\kappa = \lambda$ is stable under finite Cartesian products. As an exact corollary, an anisotropically weighted Boolean cube satisfies

$$\kappa(Q_d(w_1, \dots, w_d)) = \lambda(Q_d(w_1, \dots, w_d)) = 2 \min_i w_i.$$

This settles the rectangular-box question from the source for Boolean boxes and the cycle question for C_4 , but not for longer paths, cycles, or lattice boxes.

1 Source question and precise scope

Briestel and Caputo ask for which graphs the entropy constant $\kappa(G)$ equals the spectral gap $\lambda(G)$. They conjecture equality for all cycles of length greater than three, all paths, and rectangular boxes in $\mathbb{Z}^d$. The relevant source passage spans the following two page crops.

$$\text{perm}(A) \leq \prod_{i=1}^n \|A_i\|_{p_c}, \quad p_c = \overline{\log(n!)}, \quad (1.19)$$

where p_c is the value at which the increasing function $p \mapsto \frac{n!}{n^{n/p}}$ takes the value 1. We will see that this estimate follows from the case $\ell = n - 1$ of our Theorem 1.17, which implies a sharp subadditivity estimate for the entropy functional, see Lemma 4.1 below. The use of entropy to prove upper bounds on the permanent goes back to [46, 42]. We refer to [31, 5] for further generalizations of the Bregman-Minc theorem.

1.4. Miscellaneous remarks. We end this introduction with a few remarks on open problems and conjectures. It would be nice to compute the constant $\kappa(G)$ defined in (1.3) for various classes of graphs. Besides the complete graph $c_{xy} \equiv 1$ which is contained in Theorem 1.1, see Remark 1.2, the determination of $\kappa(G)$ remains in general a difficult problem, much as in the case of the (modified) log-Sobolev constants. In Proposition 2.10 below we consider the star graph $c_{xy} = \mathbf{1}_{x=x_0}$ for some fixed vertex $x_0 \in V$, for which we determine the entropy constant asymptotically. An interesting question is to determine for which graphs one should have $\kappa(G) = \lambda(G)$. For the log-Sobolev constant it was shown [19] that this is the case for even cycles; see also [18]. We believe that the same holds for the entropy constant. Moreover, we believe that the identity $\kappa(G) = \lambda(G)$ could extend to all n -cycles with $n > 3$, and to all paths, that is graphs defined by

$c_{xy} = \mathbf{1}_{|x-y|=1}$, $x, y \in \{1, \dots, n\}$, and more generally for rectangular boxes in $\mathbb{Z}^d$, $d \geq 2$. As we point out in Remark 2.2 below, if a graph G satisfies the identity $\kappa(G) = \lambda(G)$ then its (modified) log-Sobolev constants are also determined in terms of the spectral gap.

A very interesting question in the setting of permutations is to determine for which hypergraphs one has $\kappa[\alpha, \mathcal{S}_n] = \kappa[\alpha]$. As we have seen in Remark 1.10 this cannot hold for the mean field case. However, this could be the case for certain specific graphs such as the path or more generally for rectangular boxes in $\mathbb{Z}^d$, $d \geq 2$. As a consequence of (1.16), that would allow one to obtain sharp mixing time bounds for the interchange process on such graphs. Together with the entropic characterization of the cutoff phenomenon recently developed by Salez [43], that may even provide a direct proof of the cutoff phenomenon

The result below proves an exact tensorization theorem that resolves the Boolean-box subfamily $\{0, 1\}^d$, including C_4 . It is a substantial partial result, not a solution for general paths, cycles, or rectangular boxes.

2 Definitions and normalization

Let $G = (V, c)$ be a finite weighted undirected graph, so $c_{xy} = c_{yx} \geq 0$ and $c_{xx} = 0$. Write μ for the uniform probability measure on V . For $f : V \rightarrow [0, \infty)$, define

$$\text{Ent}_\mu(f) = \mu[f \log f] - \mu[f] \log \mu[f]$$

with the convention $0 \log 0 = 0$. For $a, b \geq 0$, put

$$\text{ent}(a, b) = \frac{a}{2} \log \frac{2a}{a+b} + \frac{b}{2} \log \frac{2b}{a+b}.$$

This is the entropy of the two-point function (a, b) under the uniform measure. In the normalization of the source, the graph entropy energy is

$$\mathcal{E}_G(f) = \frac{2}{|V|} \sum_{x, y \in V} c_{xy} \text{ent}(f(x), f(y)),$$

and

$$\kappa(G) = \inf_{\substack{f \geq 0 \\ \text{Ent}_\mu(f) > 0}} \frac{\mathcal{E}_G(f)}{\text{Ent}_\mu(f)}.$$

For another weighted graph $H = (W, d)$, the Cartesian product has vertex set $V \times W$ and conductances

$$q_{(x,u),(y,v)} = c_{xy} \mathbf{1}_{\{u=v\}} + d_{uv} \mathbf{1}_{\{x=y\}}.$$

All measures below are uniform.

3 Proof intuition

There are two independent facts. First, entropy of a function of two coordinates is no larger than the average entropy seen along the first coordinate plus the average entropy seen along the second. Second, every edge of a Cartesian product changes exactly one coordinate, so its graph entropy energy is *exactly* the sum of those same two averaged fiber energies. Applying the best factor inequality on each fiber therefore gives the lower bound for the product constant. A function depending on only one coordinate recovers the corresponding factor quotient, giving the matching upper bound.

4 Entropy tensorization

Lemma 1 (Two-coordinate entropy tensorization). *Let μ, ν be probability measures on finite sets V, W . For every $F : V \times W \rightarrow [0, \infty)$,*

$$\text{Ent}_{\mu \otimes \nu}(F) \leq \int_W \text{Ent}_{\mu}(F(\cdot, u)) d\nu(u) + \int_V \text{Ent}_{\nu}(F(x, \cdot)) d\mu(x).$$

Proof. Set $h(u) = \int_V F(x, u) d\mu(x)$. Expanding the definitions gives the entropy chain rule

$$\text{Ent}_{\mu \otimes \nu}(F) = \int_W \text{Ent}_{\mu}(F(\cdot, u)) d\nu(u) + \text{Ent}_{\nu}(h).$$

The functional $g \mapsto \text{Ent}_{\nu}(g)$ is convex on the nonnegative cone. Indeed, this is the log-sum inequality (and the boundary follows by continuity). Since $h = \int_V F(x, \cdot) d\mu(x)$, Jensen's inequality therefore yields

$$\text{Ent}_{\nu}(h) \leq \int_V \text{Ent}_{\nu}(F(x, \cdot)) d\mu(x).$$

Combining the two displays proves the assertion. $\square$

5 Exact Cartesian-product formula

Theorem 1. *For any finite weighted graphs G and H ,*

$$\kappa(G \square H) = \min\{\kappa(G), \kappa(H)\}.$$

Proof. Let $n = |V|$, $m = |W|$, and write $F^u(x) = F(x, u)$ and $F_x(u) = F(x, u)$. Splitting the ordered edge sum according to which coordinate changes gives the exact identity

$$\mathcal{E}_{G \square H}(F) = \frac{1}{m} \sum_{u \in W} \mathcal{E}_G(F^u) + \frac{1}{n} \sum_{x \in V} \mathcal{E}_H(F_x). \quad (1)$$

For example, the first term on the right is

$$\frac{1}{m} \sum_u \frac{2}{n} \sum_{x, y} c_{xy} \text{ent}(F(x, u), F(y, u)) = \frac{2}{nm} \sum_{u, x, y} c_{xy} \text{ent}(F(x, u), F(y, u)),$$

which is precisely the first-coordinate contribution to the product energy. The second term is identical in the other coordinate.

Put $k = \min\{\kappa(G), \kappa(H)\}$. Apply the defining factor inequalities to every fiber in (1), and then use the preceding lemma:

$$\begin{aligned} \mathcal{E}_{G \square H}(F) &\geq k \left\{ \frac{1}{m} \sum_u \text{Ent}_{\mu}(F^u) + \frac{1}{n} \sum_x \text{Ent}_{\nu}(F_x) \right\} \\ &\geq k \text{Ent}_{\mu \otimes \nu}(F). \end{aligned}$$

Thus $\kappa(G \square H) \geq k$.

Conversely, take $F(x, u) = f(x)$. Both its global entropy and its product energy equal the corresponding quantities for f on G ; the H -coordinate energy vanishes. Taking the infimum over f gives $\kappa(G \square H) \leq \kappa(G)$. Functions of u alone similarly give $\kappa(G \square H) \leq \kappa(H)$, proving equality. $\square$

Remark 1 (Spectral gap). *The product Laplacian is the tensor sum*

$$L_{G \square H} = L_G \otimes I + I \otimes L_H.$$

Its eigenvalues are all sums of a factor eigenvalue from G and one from H . Hence

$$\lambda(G \square H) = \min\{\lambda(G), \lambda(H)\}.$$

It follows immediately that if $\kappa = \lambda$ holds for both factors, then it holds for their Cartesian product.

6 Weighted Boolean cubes

Lemma 2. *For the two-point graph $K_2(w)$ whose sole edge has conductance $w > 0$,*

$$\kappa(K_2(w)) = \lambda(K_2(w)) = 2w.$$

Proof. For $f = (a, b)$, global entropy is $\text{ent}(a, b)$. The ordered edge sum contains the edge in both directions, while the prefactor $2/|V|$ equals one. Therefore

$$\mathcal{E}_{K_2(w)}(f) = 2w \text{ent}(a, b) = 2w \text{Ent}(f).$$

Thus $\kappa(K_2(w)) = 2w$. The weighted Laplacian matrix

$$\begin{pmatrix} w & -w \\ -w & w \end{pmatrix}$$

has eigenvalues 0 and $2w$, proving the spectral statement. □

Corollary 1. *Let*

$$Q_d(w_1, \dots, w_d) = K_2(w_1) \square \dots \square K_2(w_d), \quad w_i > 0.$$

Then

$$\boxed{\kappa(Q_d(w_1, \dots, w_d)) = \lambda(Q_d(w_1, \dots, w_d)) = 2 \min_{1 \leq i \leq d} w_i}.$$

In particular, all unweighted Boolean boxes have $\kappa = \lambda = 2$, and $C_4 = K_2 \square K_2$ satisfies the cycle conjecture.

Proof. Iterate the entropy-constant product theorem and the spectral-gap product formula, using the preceding two-point computation in each coordinate. □

7 Independent checks and novelty scope

A deterministic verifier included with this packet checks:

1. entropy tensorization on 2,000 random positive two-factor arrays;
2. the exact energy decomposition on the corresponding random weighted graph products;
3. the quotient $2w$ for 100 weighted two-point instances; and
4. the claimed inequality on 1,200 weighted cubes of dimensions one through six.

These computations are checks of normalization and implementation, not a substitute for the proof.

The run’s local indexes were searched by identifier, title, and the core product/tensorization terminology. A bounded web search found the source paper and general background, but no exact statement of the displayed product formula or weighted-cube corollary. The theorem may nevertheless be folklore; the novelty claim is therefore deliberately modest and requires expert review.

An exploratory multi-start minimization of the exact quotient on paths and cycles through ten vertices found no credible strict counterexample. That experiment is not used in the proof. The cases of longer paths, longer cycles, and boxes having any side length greater than two remain outside this packet.

Reference

A. Bristiel and P. Caputo, *Entropy inequalities for random walks and permutations*, Ann. Inst. Henri Poincaré Probab. Stat. **59** (2023), no. 4, 2280–2310; arXiv:2109.06009.

Human-review recommendation

Review as a likely valid partial result. The high-value checks are: (i) equation (1) under the source normalization; (ii) convexity of entropy in the tensorization lemma; and (iii) the factor $2w$ in the two-point graph.